

Name of Student: Darshan Sur

Scholar No.: 240030008

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Information Technology

B. Tech. End-Term Examination Nov. – 2025

Subject Name/ Code: Engg. Mathematics-III (IT- 3001)

Max. Marks: 60

Time Allotted: 3 Hours

NOTE: 1. All (6) questions are compulsory. 2. Write Answers neatly with proper Q. No.

1.	(a) Discuss the rate of convergence of Newton Raphson's Method and prove that it is quadratic.	5
	(b) Find the root of the following equation: $x^2 \cdot e^{-x/2} = 1$ in the interval $[0, 2]$, by the Secant method.	5
2.	(a) Prove the relations: (i) $\Delta = \frac{\delta^2}{2} \pm \sqrt{1 + \frac{\delta^2}{4}}$ (ii) $hD = \sinh^{-1}(\mu\delta)$, all notations are in their usual meanings.	5
	(b) Estimate the production for the year 1964 and 1966 from the following data: Year (x) : 1961 1962 1963 1964 1965 1966 1967 Production (y): 200 220 260 - 350 - 430	5
3.	(a) Apply Stirling's formula to find the value of $f(1.22)$ from the following table which gives the values of $f(x) = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-\frac{x^2}{2}} dx$ at intervals of $x = 0.5$ from $x = 0$ to 2 . x : 0.0 0.5 1.0 1.5 2.0 f(x): 0.0 0.191 0.341 0.433 0.477.	5
	(b) Compute $f'(6)$ from the following table x : 0 2 3 4 7 9 y = f(x) : 4 26 58 112 466 922 Also find maximum value of $f(x)$.	5
4.	(a) A train is moving at the speed of 30m./sec., Suddenly brakes are applied. The speed of the train per seconds is given by Time (t) : 0 5 10 15 20 25 30 35 40 45 Speed(v): 30 24 19 16 13 11 10 8 7 5 Determine the distance moved by the train in 45 seconds.	5
	(b) Employ Taylor's method to obtain approximate value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$. Compare the numerical solution	5

PTD

	obtained with the exact solution.																																											
5.	(a) R is the resistance to maintain a train at speed V ; find a law of the type $R = a + bV^2$ connecting R and V , using the following data:										5																																	
	<table><tr><td>V(miles/hour)</td><td>10</td><td>20</td><td>30</td><td>40</td><td>50</td></tr><tr><td>R(lb/ton)</td><td>8</td><td>10</td><td>15</td><td>21</td><td>30</td></tr></table>										V (miles/hour)	10	20	30	40	50	R (lb/ton)	8	10	15	21	30																						
V (miles/hour)	10	20	30	40	50																																							
R (lb/ton)	8	10	15	21	30																																							
	(b) Psychological tests of intelligence and of engineering ability were applied to 10 students. Here is a record of ungrouped data showing intelligence ratio (I.R.) and engineering ratio (E.R.). Calculate the co-efficient of correlation.										5																																	
	<table><tr><td>Student</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>G</td><td>H</td><td>I</td><td>J</td></tr><tr><td>I.R.</td><td>105</td><td>104</td><td>102</td><td>101</td><td>100</td><td>99</td><td>98</td><td>96</td><td>93</td><td>92</td></tr><tr><td>E.R.</td><td>101</td><td>103</td><td>100</td><td>98</td><td>95</td><td>96</td><td>104</td><td>92</td><td>97</td><td>94</td></tr></table>										Student	A	B	C	D	E	F	G	H	I	J	I.R.	105	104	102	101	100	99	98	96	93	92	E.R.	101	103	100	98	95	96	104	92	97	94	
Student	A	B	C	D	E	F	G	H	I	J																																		
I.R.	105	104	102	101	100	99	98	96	93	92																																		
E.R.	101	103	100	98	95	96	104	92	97	94																																		
6.	(a) If $\beta_1 = \frac{1}{36}$ and $\beta_2 = \frac{35}{12}$, then find the corresponding Binomial distribution.										5																																	
	(b) Fit a Normal Curve of the following distribution.																																											
	x: 2 4 6 8 10										5																																	
	y: 1 4 6 4 1																																											
	Given that $Z(2.5) = 0.4938$, $Z(1.5) = 0.4332$, $Z(0.5) = 0.1915$.																																											

All the Best

e by

or
is
g
e